

广义若当区域上共形同胚的连续延拓问题

To Generalize Osgood-Taylor-Carathéodory Theorem

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什么是(广义)若当区域(Jordan Domain)

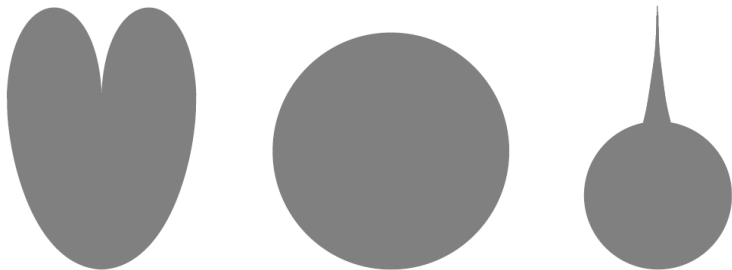


FIGURE 1. A non-convex fat domain, a disk, and a non-fat domain.

Definition (广义若当区域)

A generalized Jordan domain is a domain $U \subset \hat{\mathbb{C}}$ with the following properties:

- (a) ∂U is a Peano compactum,
- (b) each component of ∂U is either a point or a Jordan curve.

什么是广义若当区域 (Generalized Jordan Domain)

Theorem 2.1. *The following statements are equivalent for any domain $U \subset \hat{\mathbb{C}}$:*

- (1) *U is a generalized Jordan domain,*
- (2) *U is simply connected at the boundary,*
- (3) *for any continuous map f of U into a compact metric space X all cluster sets $C(f, z)$ with $z \in \partial U$ are connected.*

$$C(f, z_0) := \bigcap_{r>0} \overline{f(D_r(z_0) \cap U)}.$$

Definition 2.2. A domain $U \subset \hat{\mathbb{C}}$ is said to be **simply connected at a point** $z_0 \in \partial U$ provided that for any disk $D_r(z_0)$ centered at z_0 with radius $r > 0$ there is a set N_r open in $\hat{\mathbb{C}}$ with $z_0 \in N_r \subset D_r(z_0)$ such that $N_r \cap U$ is connected.

经典的Caratheodory连续延拓定理(continuity theorem)

Theorem A (Continuity Theorem). *A conformal homeomorphism $\varphi : \mathbb{D} \rightarrow \Omega \subset \hat{\mathbb{C}}$ of the unit disk $\mathbb{D} = \{z : |z| < 1\}$ has a continuous extension $\bar{\varphi} : \bar{\mathbb{D}} \rightarrow \bar{\Omega}$ if and only if $\partial\Omega$ is locally connected.*

Theorem B (OTC Theorem). *A conformal homeomorphism $\varphi : \mathbb{D} \rightarrow \Omega \subset \hat{\mathbb{C}}$ has a continuous and injective extension to $\bar{\mathbb{D}}$ if and only if the boundary $\partial\Omega$ is a Jordan curve.*

Theorem C (OTC Theorem—for Countably Connected Circle Domains). *Every conformal homeomorphism $\varphi : D \rightarrow \Omega$ of a countably connected circle domain D onto a circle domain Ω extends to be a homeomorphism between $\bar{D}$ and $\bar{\Omega}$.*

推广的Caratheodory连续延拓定理(continuity theorem)

Theorem D ([15, Theorem 2.1]). Let D be a circle domain in $\hat{\mathbb{C}}$ whose boundary has σ -finite linear measure. Let Ω be another circle domain and let $\varphi : D \rightarrow \Omega$ be a conformal homeomorphism. Then φ extends to be a homeomorphism $\bar{\varphi} : \bar{D} \rightarrow \bar{\Omega}$.

Theorem E ([25, Theorem 6.1]). Let D be a circle domain with $\infty \in D$ and let h be a conformal map from D onto another domain Ω with $\infty = h(\infty) \in \Omega$. Suppose that D satisfies the q -quasihyperbolic condition and that the complementary components of Ω are points and uniformly fat closed Jordan domains.

Then h extends to be a homeomorphism from $\bar{D}$ onto $\bar{\Omega}$.

Theorem F (OTC Theorem—for Cofat Generalized Jordan Domains). Let $\varphi : \Omega_1 \rightarrow \Omega_2$ be a conformal homeomorphism between generalized Jordan domains that are countably connected. If Ω_1 and Ω_2 are both cofat then φ extends to a homeomorphism between $\bar{\Omega}_1$ and $\bar{\Omega}_2$.

推广的连续延拓定理—3个前提

Extension Problem. Under what conditions does a conformal homeomorphism $\varphi : U \rightarrow \Omega \subset \hat{\mathbb{C}}$ of a circle domain (or generalized Jordan domain) have a continuous extension $\bar{\varphi} : \bar{U} \rightarrow \bar{\Omega}$?

Assumption 1 The boundary $\partial\Omega$ is a Peano compactum.

Assumption 2 diameters of nondegenerate components of $\partial\Omega$ have finite sum

Assumption 3 point components of ∂U form a set of sigma-finite linear measure.

Assumption 3* point components of $\partial\Omega$ form a set of sigma-finite linear measure.

推广的连续延拓定理—2个版本

Theorem 1 (First Generalization of Continuity Theorem).

Let $\Omega \subset \hat{\mathbb{C}}$ be a domain satisfying Assumptions 1–2.

Let $D \subset \hat{\mathbb{C}}$ be a circle domain satisfying Assumption 3.

Then any conformal homeom $\varphi: D \rightarrow \Omega$ has continuous extension $\overline{\varphi}: \overline{D} \rightarrow \overline{\Omega}$.

Theorem 2 (Second Generalization of Continuity Theorem).

Let $\Omega \subset \hat{\mathbb{C}}$ be a domain satisfying Assumptions 1–2 and 3.*

Then any conformal homeomorphism $\varphi: D \rightarrow \Omega$ of a circle domain D onto Ω has a continuous extension $\overline{\varphi}: \overline{D} \rightarrow \overline{\Omega}$.

推广的OTC定理——(定理1/2的推论)

Theorem 3 (Generalized OTC Theorem). *Let Ω_1, Ω_2 be generalized Jordan domains. Let $\{P_n\}$ (respectively, $\{Q_n\}$) denote the non-degenerate components of $\partial\Omega_1$ (respectively, of $\partial\Omega_2$). Then any conformal homeomorphism $f : \Omega_1 \rightarrow \Omega_2$ extends to a continuous map from $\overline{\Omega_1}$ onto $\overline{\Omega_2}$ provided that the following two conditions are both satisfied:*

- (i) *The inequality $\sum_n \text{diam}(Q_n) < \infty$ holds for the diameters of Q_n .*
- (ii) *The point components of $\partial\Omega_1$ or $\partial\Omega_2$ form a set of σ -finite linear measure*

If, in addition, $\sum_n \text{diam}(P_n) < \infty$ then f extends to a homeomorphism from $\overline{\Omega_1}$ onto $\overline{\Omega_2}$.

Consequently,

this extension further extends to a homeomorphism of $\hat{\mathbb{C}}$ onto itself.

推广 Whyburn 的一个经典定理

Theorem 4. *Let $\Omega \subset \hat{\mathbb{C}}$ be a domain. The following are equivalent:*

(1) $\partial\Omega$ is a Peano compactum;

(2) Ω has Property S;

(3) Every point of $\partial\Omega$ is locally accessible;

(4) Every point of $\partial\Omega$ is locally sequentially accessible;

(5) Ω is finitely connected at the boundary;

(6) The completion of Ω under the Mazurkiewicz distance d is compact.

前提2 “不可缺少” ——基于“超边界极值长度”的例子

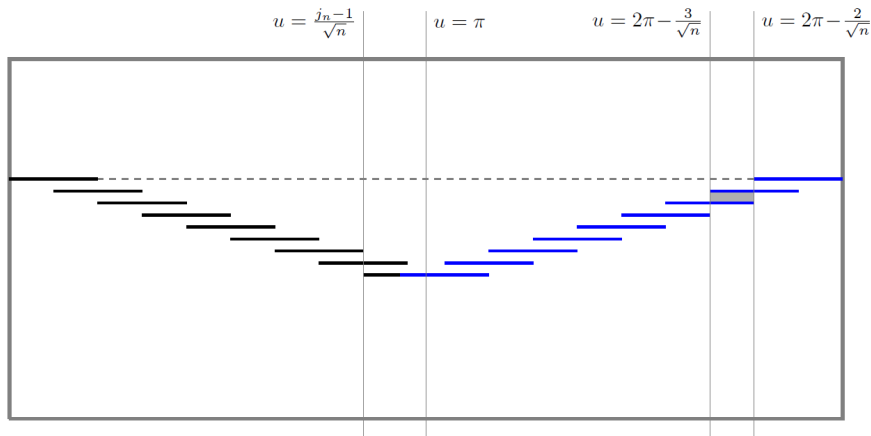


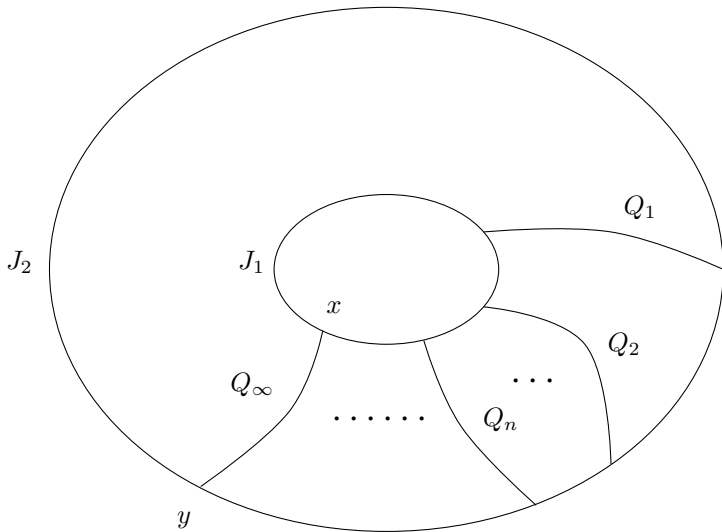
FIGURE 6. The segments $\alpha_{n,t}$ and how they are distributed, with $n=9=j_n$.

- [29] Oded Schramm, *Transboundary extremal length*, J. Anal. Math. **66** (1995), 307–329, DOI 10.1007/BF02788827. MR1370355

THANK YOU ALL

♣ With more time, we may describe $\mathcal{D}_K$ and the origin of Peano compactum.

Let us start from [Schönflies relation](#), defined on [all planar compacta](#).



Criterion for Peano Compacta in the Plane

Theorem (1. Based on a Schönflies result [Kur68, p.515])

Every *locally connected or finitely suslinian* compactum $K \subset \mathbb{C}$ satisfies Schönflies condition: every fiber of R_K is a singleton.

Theorem (2. Nontrivial Characterization for Schönflies Condition)

A compactum $K \subset \mathbb{C}$ satisfies the Schönflies condition if and only if:

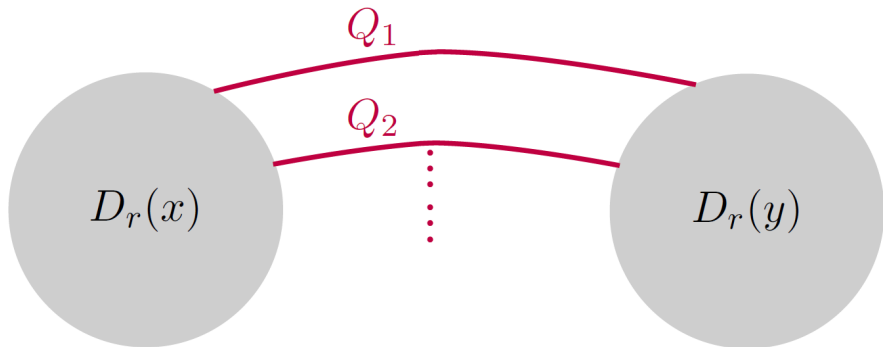
- (1) every of its components is locally connected;
- (2) at most *finitely many components of K*
have a diameter *greater than* any fixed constant $C > 0$.

Therefore $K \subset \mathbb{C}$ satisfies *Schonflies condition* $\Leftrightarrow$ it is a *Peano compactum*.

- ♣ The notion of *Peano compactum* extends that of *Peano continuum*, so that
A *Cantor Set* is a Peano compactum.
A *Peano Continuum* means a *connected* Peano compactum.

- ♣ If we have more time, let us look at the *closed relation* $\overline{R_K}$.

The closure of R_K for compact $K \subset \hat{\mathbb{C}}$



for any $x \in K$

the $\overline{R_K}[x] = \{y : (x, y) \in \overline{R_K}\}$ at x is connected.



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Richard Feynman

ATOMIC HYPOTHESIS in PHYSICS

1. All things are made of **atoms**;
2. **(Atoms are)** little particles that move around in perpetual motion,
3. Attracting each other when they are a little distance apart,
4. But repelling upon being squeezed into one another.

♣ From this we may **recover** a large part of **modern physics** !

Hahn-Mazurkiewicz

Moore, Whyburn, ...

FitzGerald-Swingle

Thurston, Kiwi, ...

Blokh-Curry-Oversteegen

Loridant-Luo-Yang

Luo-Yang-Yao

Feng-Luo-Yao,

THEORY OF ATOMS for Planar Compacta

1. Every compactum $K \subset \hat{\mathbb{C}}$ is made of **atoms**
2. $\mathcal{D}_K = \{\text{atoms}\}$ is a monotone decomposition of K
s.t. (1) the **quotient $\mathcal{D}_K$** is a **Peano compactum**.
(2) $\mathcal{D}_K$ **refines** every other such decomposition.
3. $h(\mathcal{D}_K) = \mathcal{D}_{h(K)}$ for any **embedding** $h : K \rightarrow \hat{\mathbb{C}}$
Here $h(\mathcal{D}_K) := \{h(d) : d \in \mathcal{D}_K\}$.
4. $f^{-1}(\mathcal{D}_K) = \mathcal{D}_{f^{-1}(K)}$ for any **rational** $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$
Here $f^{-1}(\mathcal{D}_K) := \{\text{components of } f^{-1}(d) : d \in \mathcal{D}_K\}$.

♣ We want to re-interpret or to recover, to extend or to upgrade
Theorems/Results/Questions that are related to Peano continua.